

Dynamic Analysis of Neural Populations using Mean Field Theory

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Introduction. The brain's complex organization - from microscale to macroscale - presents unique challenges in studying its dynamics. Mean Field theory offers a powerful framework to address this complexity by approximating the behavior of large neural populations through their average activity. By leveraging assumptions of weak coupling and large population sizes, this approach enables the use of statistical techniques to describe the population dynamics accurately.

1 Methods

In our analysis, we will investigate the system's behavior through various visualizations: activation functions for excitatory and inhibitory populations, nullclines in the phase plane, vector fields, and trajectories for different initial conditions. We will explore how key parameters - including the time constant ratio, input current, and theta threshold value - affect these dynamics. By examining fixed points, nullclines, and phase plane trajectories, we aim to understand how these fundamental parameters shape the dynamical properties of the neural populations and their interactions.

2 Aim

In our study, we use mean field theory [1] to analyze the large-scale behavior of neural populations by simplifying complex neural networks into more manageable models. Instead of focusing on individual neurons and their interactions, we consider entire populations and approximate their collective dynamics using averaged interactions.

In particular, we focus on the Wilson-Cowan model [3] which describes the system through two coupled differential equations: one governing the firing rate of excitatory neurons and the other for inhibitory neurons.

$$\tau_E \frac{dr_E}{dt} = -r_E + F_E(w_{EE}r_E - w_{EI}r_I + I_E) \quad (1)$$

$$\tau_I \frac{dr_I}{dt} = -r_I + F_I(w_{IE}r_E - w_{II}r_I + I_I) \quad (2)$$

Both equations include an exponential decay term, representing the natural decrease in firing rate over time. Additionally, each population has a transfer function that determines how its activity evolves, with potentially different forms for excitatory and inhibitory neurons.

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In the equation for the excitatory population, the first term of the transfer function accounts for excitatory recurrence, meaning that excitatory neurons reinforce each other’s activity based on their synaptic connections. However, this activity is counterbalanced by inhibition: as excitatory neurons fire more, they activate inhibitory neurons, which in turn suppress excitatory activity. This interplay creates a dynamic equilibrium.

Another key factor is the external current input, which directly influences the activity of the excitatory population. Our final goal is to observe how the system’s dynamics change by adjusting different parameters—such as the strength of synaptic connections, the balance between excitatory and inhibitory networks, the intensity of external input, and the activation threshold. These parameters shape the interaction between excitatory and inhibitory populations, ultimately leading to a form of self-regulation that stabilizes neural activity.

3 Methods

3.1 Weights

In this analysis, we explore how varying the synaptic weights w_{EE} , w_{IE} , w_{EI} , and w_{II} affects the dynamics of the Wilson-Cowan model. The left panel shows the excitatory (blue) and inhibitory (red) nullclines, whose intersections represent the fixed points of the system. When these synaptic weights are adjusted, the effective coupling between the excitatory and inhibitory populations changes, shifting the nullclines and, consequently, the locations of their intersections. This directly alters the system’s equilibrium structure.

Using interactive sliders to modify the weights allows us to dynamically observe how the nullclines move. The right panels illustrate the temporal evolution of excitatory and inhibitory activities under two different initial conditions. In one case, the activities rapidly decay to low levels, while in the other, they stabilize at higher rates, indicating a balanced E/I interaction.

Mathematically, these synaptic weights adjust the coupling terms in the Wilson-Cowan equations, so even small changes can lead to significant differences in network dynamics and stability. This interactive approach provides continuous insight into how the network transitions from one state to another, making it a powerful tool for understanding the role of synaptic weights in neural dynamics.

3.2 Input Current

When the input current is increased, the net input becomes larger, effectively shifting the excitatory nullcline. This shift alters the location where the excitatory nullcline intersects with the inhibitory nullcline, which determines the fixed points of the system.

This is important because it allows us to see, in a continuous manner, how the network transitions from one state to another—for example, from a low-activity state to a high-activity state.

Mathematically, the input current enters the excitatory equation as an additive term, so when it increases, the overall input to the excitatory population is effectively raised. This causes the curve representing the condition $dr_E/dt = 0$ to move upward in the phase plane.

What this means in practice is that for any given level of inhibitory activity, the excitatory neurons will now reach a higher firing rate at equilibrium.

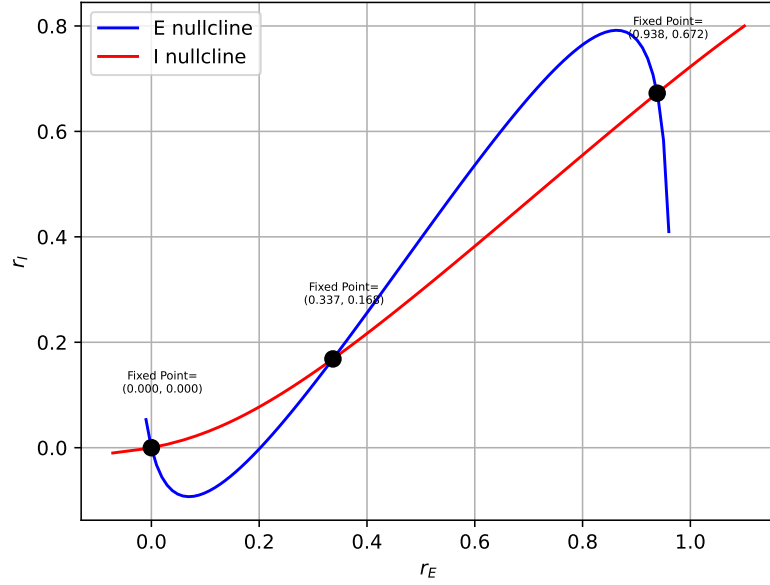


Figure 1: Nullclines of excitatory and inhibitory populations with computed fixed points.[2]

3.3 Time Constant

The inhibitory time constant τ_I determines how quickly the inhibitory population responds to changes in excitatory activity, influencing both the system's dynamics and its equilibrium behavior. When τ_I is small, inhibition reacts rapidly, leading to faster stabilization of the network. On the other hand, a larger τ_I results in slower inhibitory feedback, which can cause prolonged transients or oscillatory behavior before reaching equilibrium.

The left panel of the figure illustrates how different values of τ_I affect the phase-plane structure, including the position of the nullclines, the fixed point, and the trajectories of excitatory and inhibitory activity. The vector field provides insight into the directional flow of activity, showing how the system evolves over time. The right panel presents the temporal evolution of excitatory (r_E , blue) and inhibitory (r_I , red) firing rates, highlighting how the network stabilizes after an initial transient phase.

By interactively adjusting τ_I , we can observe how the system transitions between different dynamical regimes. A faster inhibitory response helps the network quickly reach a steady state, while a slower inhibitory timescale can delay stabilization, allowing more complex temporal dynamics to emerge. Understanding this balance between excitation and inhibition is essential for characterizing the stability and flexibility of neural networks.

3.4 E/I Balance

The analysis explores how the balance between excitatory (E) and inhibitory (I) interactions influences system stability by varying the synaptic weights w_{EI} (inhibitory effect on excitatory neurons) and w_{IE} (excitatory effect on inhibitory neurons) within a specified range.

Two heatmaps were generated to visualize the impact of these parameters on system stability.

The first heatmap, titled "Stability HeatMap: Max Real Eigenvalue of Jacobian," displays the maximum real part of the eigenvalues of the Jacobian matrix at the fixed points. The map reveals a transition from a uniformly red region on the left to a uniformly blue region on the right, indicating

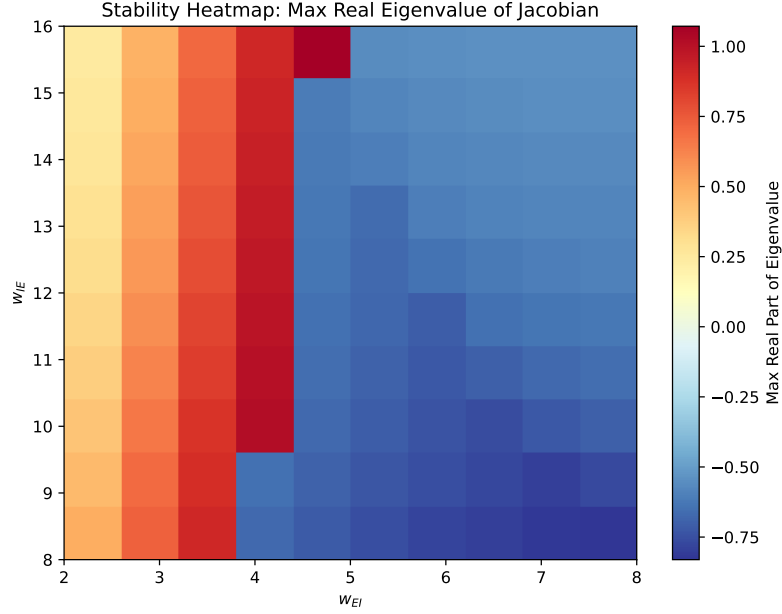


Figure 2: Stability heatmap: Maximum real eigenvalue of the Jacobian as a function of inhibitory and excitatory synaptic weights, indicating network stability.[2]

a shift in system stability across the parameter space. This suggests that changes in the balance of inhibitory and excitatory synaptic weights affect the system's stability in a measurable way, with certain combinations of w_{EI} and w_{IE} yielding more stable fixed points.

The second heatmap, "Stability Map of E/I Balance," is uniformly colored in aqua green, which indicates no significant change in stability across the entire range of parameters. This uniformity suggests that, within the specified ranges, variations in the excitatory and inhibitory synaptic weights do not lead to qualitative changes in the system's dynamical behavior or stability.

These findings are important because they demonstrate that, in the regime tested, the system maintains a stable equilibrium, which is crucial for avoiding extreme conditions such as excessive inhibition or runaway excitation.

3.5 Activation Threshold

We want to analyze the system's stability under different parameter settings of the threshold activation functions. To do so we plotted a heat map which illustrates how the maximum real part of the eigenvalues of the Jacobian of the coupled differential equations varies as a function of the activation thresholds θ_E (for excitatory neurons) and θ_I (for inhibitory neurons).

Regions in warm colors (red, orange, yellow) correspond to positive real eigenvalues, indicating that the system is unstable in those conditions. Conversely, cool colors (blue, light blue) represent negative real eigenvalues, meaning the system is stable as activity returns to the fixed point over time. The transition between these two regions marks the boundary between stability and instability.

From the heatmap, we can observe that at low values of θ_E and θ_I (left side of the graph), the system tends to be unstable, suggesting that neurons become overly sensitive and react excessively to inputs. As we increase either θ_E or θ_I , the system transitions towards stability, as indicated by

the gradual shift to cooler colors. This implies that higher activation thresholds require stronger inputs to trigger neuronal activity, preventing excessive firing and promoting stability.

Overall, this analysis highlights the importance of the balance between excitation and inhibition. If the activation thresholds are too low, excitatory neurons may over-activate, leading to instability. On the other hand, sufficiently high thresholds ensure that the system remains in a stable state, avoiding uncontrolled fluctuations in neural activity.

4 Discussion

In conclusion, our study shows that using mean field theory with the Wilson-Cowan model is a powerful way to understand how large groups of neurons behave. By changing key parameters like synaptic weights, input current, time constants, and activation thresholds we demonstrated how these factors adjust the balance between excitation and inhibition, shifting the nullclines and altering the fixed points of the system. The interactive visualizations let us see these changes in real time and highlight that even small adjustments can have significant effects. Overall, our findings reveal the strong and flexible nature of neural networks, providing a solid basis for future research into more detailed and realistic models. In the context of epilepsy, this model has been used to investigate how shifts in the excitation-inhibition balance can lead to seizure-like activity. Specifically, the model illustrates how transitions between normal and pathological brain states can occur through bifurcations, where small parameter changes result in sudden shifts to a hyperexcitable regime. This transition mirrors the abrupt onset of seizures observed in clinical settings.

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